A Comparative Study of VAEs and GANs for Generative Modeling on CIFAR-10

- Summary Report
- Based on the document provided

Research Objectives

- Comparative analysis of VAEs and GANs as generative models.
- Evaluation on the CIFAR-10 dataset.
- Focus on sample quality, latent representations, and training dynamics.
- Understanding how these models learn representations and generate data.
- Informing practical trade-offs when deploying such models.
- Investigation of β-VAE and WGAN-GP architectures for improved interpretability and stability.

Experimental Methodologies

- Models Studied: VAE, β-VAE, GAN, WGAN-GP.
- Architectural structures and loss functions were detailed in the full report.
- VAE: Convolutional encoder and transposed convolutional decoder with reparameterization trick.
- GAN: Adversarial training between a generator (transposed convolutions) and a discriminator.
- WGAN-GP: Wasserstein GAN with Gradient Penalty.

Key Findings and Conclusions

- β-VAE: Enforces latent disentanglement, improving interpretability but potentially compromising reconstruction fidelity.
- GANs: Generate high-quality, sharp, and realistic samples due to their adversarial objective. However, they are difficult to train, control, prone to instability, and have less interpretable latent spaces.
- WGAN-GP: Improves training stability and mitigates mode collapse compared to standard GANs. In this experiment, its performance based on FID score was not significantly better than simpler GANs.
- FID Limitation: FID alone is insufficient for judging a model's overall utility. Qualitative aspects are essential.
- Model Suitability: GANs excel in photorealistic sample generation, while VAEs and β-VAEs are better for interpretability and structured latent exploration.

Suggestions for Future Work

- Exploring hybrid models, such as VAE-GANs, to combine the strengths of both paradigms.
- Investigating the use of richer priors within the VAE framework to potentially enhance generative capabilities and sample quality.